Citation Equity: Which GEO Writing Advice Moves Citation? A Tool Built From the Answer

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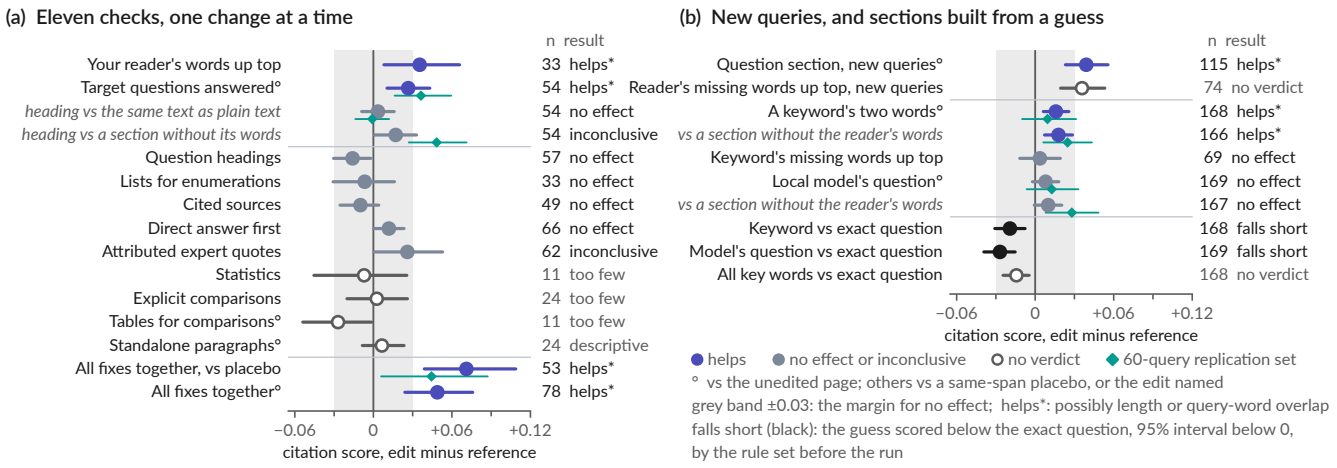


Figure 1: Of eleven checks, only the reader's own words near the top clearly moved citation; the gain held on new queries, and a keyword helped less. Difference in citation score from a placebo that rewords the same spans, or from the unedited page (°), 95% intervals; n: queries where the check applied and its edit (and placebo) passed validation; helps is decided by the corrected test before the ±0.03 band is read, so a significant row can sit inside it. (a) Each check alone on 80 GEO-bench queries (260 for checks with too few pages at 80); two italic rows test whether Target questions answered, the question section, gains from its heading or its words. (b) The question section on 180 new queries; then the section built from a keyword or a local model's question, over all 260 queries; the bottom rows compare each guess, and a section with all key words, with the exact question.

ABSTRACT

Writers are told that question headings, lists, statistics, quotes and cited sources help AI answer engines cite their pages. We tested ten such checks and one of our own by editing real web pages one change at a time, each against a paraphrase of the same spans or the unedited page, while open models answered from fixed sources. Only one kind of single change clearly raised citation: the reader's own words near the top of the page, by about 0.03 on a 0-to-1 citation score. The heading form did not matter. Question headings, lists, cited sources and a direct answer first showed no effect larger than 0.03. On queries the first test never used, the gain held (+0.039). A keyword's two words helped, but less. A question a local model wrote from that keyword showed no effect larger than 0.03. We present Citation Equity, an open-source writing tool rebuilt around this result. In the browser and offline, it starts from the questions readers ask, shows which of their words a draft lacks near the top, flags settings that keep a page out of answers, and labels the other advice with what our tests found.

CCS CONCEPTS

• **Information systems** → **Web searching and information discovery**; • **Human-centered computing** → *Interactive systems and tools.*

KEYWORDS

generative engine optimization, answer engines, citation, writing tools, controlled tests, null results

1 INTRODUCTION

Answer engines, systems that answer in prose and cite web sources, now decide which pages a reader sees cited. In a field experiment, a search page with only AI Mode cut click-through by 18.8 percentage points [10]. Writers respond with generative engine optimization (GEO): advice and products for getting a page cited.

GEO-bench and C-SEO Bench tested content rewrites (statistics, quotes, citations; keyword stuffing in GEO-bench) [2, 7]; the structural advice (question headings, lists, tables, an answer first) had been tested only as a formatting bundle, with no consistent effect [8], or against the unedited page [7], not one change at a time against a same-length placebo. The best-known figure, gains of up to 40%, comes chiefly from one setup with five sources in one model's prompt [2]; a survey of 45 studies rejects it as a general claim [6], and adding statistics lowered citation rank in 19 of 24 C-SEO Bench settings [7]. Of 24 commercial products whose public pages we read, most sell brand tracking and seven also rewrite the text,[1] yet writers feel less ownership of text an AI writes for them [3].

We built that advice into a writing tool and tested each of its ten checks on its own, beside candidate checks of our own; the one candidate that helped became its eleventh. For each check, an open model or a fixed rule made that one edit to real web pages, and open models answered from fixed sources (80 GEO-bench queries, with 60 from C-SEO Bench as a replication set); every test was planned in advance. Only the reader's own words near the top of the page clearly raised citation (Figure 1a). A follow-up found that the heading itself added nothing. A last test asked what a writer can know: on the 180 new queries (115 where the edit applied) the gain held, a keyword's two words helped, but less, and a question a local model wrote from the keyword showed no effect larger than 0.03 (Figure 1b).

We present *Citation Equity*, an open-source writing tool rebuilt around these results (Figure 2).[2] It starts from the questions readers ask, names the words a draft lacks near the top without inserting them, and labels the rest of the advice with what our tests found. The equity is of access, not of rankings: the tool is free, open and offline, so the evidence is not reserved for brands that pay for tracking tools. Our contributions are (1) a controlled test of ten GEO checks and one of our own, with placebos, a replication set, an editor swap, new queries and guessed questions; (2) the finding that the reader's vocabulary, not any formatting check we could judge, clearly moved citation; and (3) an open, offline writing tool rebuilt from that finding, with a live demo, code and data.

2 WHAT WE TESTED

The test. For a query and a check, an open model (Gemma-4-31B-it) edits only the spans the check flags, and validators reject an edit that adds new numbers, links or names, or misses the check. The reference rewords the same spans, at the same length for a rewrite and with nothing inserted for an additive check, or is the unedited page (° in Figure 1). Two open judges (Gemma-4-12B-it and Qwen3.8-27B) answer the query from 4 sources, 8 times each with the page in each slot twice, and the page's *citation score* is $(5 - r)/4$ when it is the r-th source cited, else 0. A result needs 30 or more queries, and tests are Holm-corrected within their family. "No effect larger than 0.03" means the 90% interval lies inside ±0.03, so smaller effects stay possible; contrasts the plan left unjudged read *no verdict* (plans, code and data: footnote 2). Each contrast uses the pages where the edit, and its placebo when it has one, passed the validators; several model-edited checks lost over 20% of their pairs to them (*Your reader's words up top* kept 33 of 63 applicable pages), so the estimates hold for pages where the edits were clean.

Eleven checks. Of single changes, only the two that add the reader's own words near the top clearly raised the score (Figure 1a). *Your reader's words up top*, a candidate check the tool added because it helped, works the question's missing words into the title and opening (+0.035 against its placebo; replication: too few pages). *Target questions answered* was tested, by rule, with the reader's question as a heading under the title over a copy of the page's best-matching sentence: the *question section* (+0.027 against the unedited page, 95% CI [+0.010, +0.043], Holm $p = 0.006$). All fixes together, which include the question section, also helped (+0.049). The heading form added nothing: the question as a heading did no better than the same text set as plain (+0.003). Against a same-size section without the reader's words, the question section helped on the replication set (+0.048) and was inconclusive on the main set (+0.017). Re-edited by another model with one judge swapped, the reader's-words gain kept its sign (+0.029) on 29 queries, one short of our minimum. A follow-up added 180 queries and one more editing attempt for the checks with too few pages: over the 260, question headings, cited sources and a direct answer first showed no effect larger than 0.03, as did lists over the first 80 and a candidate definition sentence under the title; quotes were inconclusive, and three checks still had too few pages (Figure 1a).

When the writer has to guess. The gain held on new queries, and a keyword helped, but less. Both gains above used the reader's exact question, which a writer rarely sees, so a last test reran the question section on 180 new queries, which the first test never used, and then built the section from two guesses the tool offers: a keyword, here the question's two rarest key words, and a question the same editor model wrote from it with the tool's prompt; answers were still judged against the true question. The question section helped on the new queries (+0.039, 95% CI [+0.023, +0.056], $n = 115$), and the reader's missing words up top held there too (+0.036 against their placebo, $n = 74$, no verdict). Over all 260 queries the keyword helped against the unedited page (+0.016) and against a section without the reader's words (+0.018; Holm $p = 0.006$ for both); the first gain rests mainly on the second judge (Qwen3.8-27B +0.021; Gemma-4-12B alone +0.010, not significant). Yet the exact question beat it by +0.019, 95% CI [+0.008, +0.031]: by the rule set before the run, the guess *falls short of the exact question.* On the new queries alone the keyword kept about two thirds of the gain (90% CI 0.41

[1] For example https://www.evertune.ai/pricing and https://writesonic.com/pricing, opened 24 October 2026.

[2] Live demo: https://shehral.github.io/citation-equity/. Code (MIT): https://github.com/shehral/citation-equity. Data (all tests): https://huggingface.co/datasets/Shehral/citation-equity-experiments. Research page: https://shehral.github.io/citation-equity/research/. Video: https://youtu.be/kbU9t3ewVUk. **[OWNER:** Re-record for the question-first tool.]

to 0.93; no verdict). Part of the edge is the query string itself, which the judge's prompt also shows (Figure 1b, bottom row, no verdict). The model's question showed no effect larger than 0.03, and neither did the keyword's missing words up top against their placebo.

> Each main-set gain carries the plan's note, possibly length or query-word overlap, but length alone did not do it: the reader's words beat same-length controls (+0.035; a keyword +0.018). For writers that is the point: the formatting they are sold did not clearly move citation, and the reader's vocabulary did.

3 CITATION EQUITY

From a score to questions. The tool shows no total: it starts from the reader's questions, counts which ones a draft covers, and labels every check with what Section 2 found (Figure 2). It replaces a 0-to-100 checklist score: on two sample drafts, the old tool's suggestions raised that score by 26 to 29 points while the drafts' share of citations among eight sources rose by 1 to 3 points from about 17 (an exploratory run on four open models).

The flow. Your questions asks "What will your readers ask?": the writer types questions, pastes keywords, imports a Search Console or Bing Webmaster export, or loads an example. *Your draft* leads with "Questions covered: N of M". A question is covered when its key words appear in the title, first heading or first paragraph, and a section opens with a plain sentence that uses most of them. The card names the missing words and never inserts them. Below sit one ready edit, the *Quick edit*, and the other checks as collapsed writing notes. *Your site* ("Can engines reach it?") checks pasted HTML and robots.txt for three blockers (a blocked AI crawler, a noindex rule, text that needs JavaScript), which rest on the engines' own documentation; the other page items are notes. *Results* ("Did it change anything?") runs the citation test on the draft before and after, and tallies whom AI answers cite when the writer pastes them.

Labels, not scores. Each check carries one fixed label for what our tests found: *Helped in our test*; *No effect in our test*, for no effect larger than 0.03; or *Unclear*, for one that ran without a verdict, since *Not tested* would be untrue. *Not tested* is kept for inputs no test judged, such as questions made from the page itself, only a placebo in our last test. The two question checks read Helped for the reader's own questions, typed or imported, though our tests used only full questions. Question headings, lists, cited sources and a direct answer first read No effect; the other five read Unclear. The *Questions covered* count asks for both measured parts at once, the words up top and an opening sentence of 8 to 40 words that answers, a stricter bar than the question section met (27% of pages): each part was tested on its own, and the count makes the writer do both, so it is a to-do list built from what helped, not a predictor. A keyword's question reads *May help less* for its answer and *No effect* for its words, and a local model's question *No effect*, measured with the 31B editor model, not the tool's 3B default; both are rules chosen after the results (Research page, footnote 2).

Evidence. Apart from our labels, the Research page grades each check on the nine studies the tool cites under a strict rule: a null or negative result in any study, even in one subgroup, makes it *mixed*.

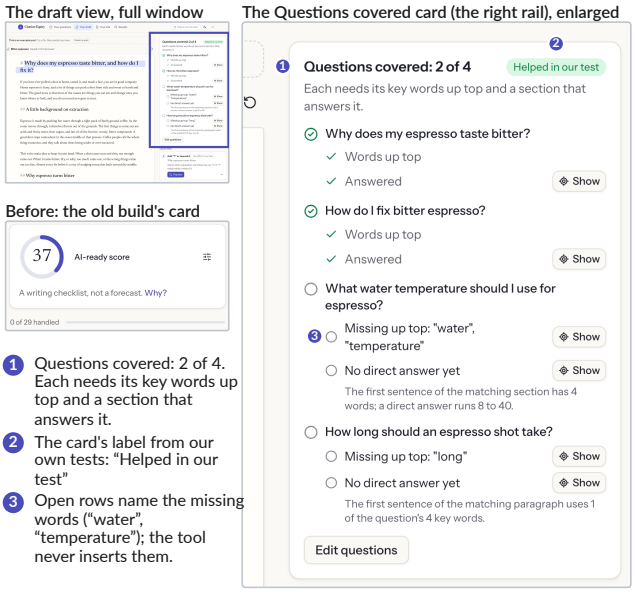


Figure 2: The draft view counts which reader questions the draft covers and names the missing words; the old build led with a checklist score. Espresso example after two typed edits, before the Quick edit; the answering sentence sits below the frame.

By that rule, coded by one author, nine checks are mixed and two untested.

Suggest, never rewrite. Comments anchored to spans of text gave writers more agency than a chat interface in a lab study [5], so every suggestion is anchored to an exact range of the draft and follows the author's edits. An edit is previewed and applied only if its text is unchanged, every action has Undo, there is no "apply all", and an optional local model proposes wording only when asked.

Architecture and the citation test. A TypeScript monorepo holds an *engine* of pure check functions (browser and CLI), a *bench* with the citation test, and a *workspace*, a static export on GitHub Pages with no server; a browser test asserts that no request leaves the origin on load. For each question the bench retrieves the top 8 of the draft and its competitor pages with BM25, and a local model (`llama3.2:3b` through Ollama, the default) answers with numbered citations under GEO's prompt [2], 8 times with shuffled source orders [6, Sec. 11.2]. Reports show the chance level and say *"It does not predict live AI engines."* With a sample's 4 questions the smallest p is 0.125, so a laptop run shows only a direction.

4 DEMONSTRATION AT THE VENUE

One laptop serves the static build and runs Ollama (`llama3.2:3b`), so the core demonstration needs no venue network; a second screen shows the hosted demo and a poster carries Figures 1 and 2.

Station 1, the 90-second loop. The visitor clicks *Try it with an example* and sees four reader questions, a draft that covers none

of them, and one blocker on the example page. The ready edit adds a question mark to a heading, common GEO advice, and its chip reads *No effect in our test*; applied through Preview, it leaves the count at 0 of 4. The visitor then opens the example's recorded before-and-after test beside its chance line: its after draft followed the old checklist and rose within noise (cited in 56% then 66% of answers; share 0.09 then 0.14 against 0.125, $p = 0.25$). A live run (64 answers, 15 minutes) fills in during each session. Last, two typed edits, a few of the reader's words in the title and one answering sentence, take the count to 2 of 4 (Figure 2).

Station 2, your readers' questions (offline). Visitors type their questions or keywords, paste their own paragraph, and see which questions it covers and which words are missing; the text never leaves the laptop.

Station 3, can engines reach your site? (online). The command-line `ce site` checks the visitor's URL for blockers over a phone hotspot; offline, pasted HTML and robots.txt stand in.

5 RELATED WORK

The bench reuses GEO's prompt and word share [2]; our tests follow C-SEO Bench and What Gets Cited, which edit one page and hold the rest fixed [7, 8]. Our vocabulary finding fits earlier ones: relevance edits moved models and stylistic edits did not [9]; a page holding the query's terms won the first citation in four of six models, and formatting-only matched pairs had no consistent effect [8]. GEO's keyword stuffing added query keywords anywhere in the page and did not help [2]; our edit adds only the missing words, to the title and opening. A preprint that restructured 200 articles (headings, chunking, lists and tables at once) reports more citations across six engines [11]; all our fixes together also helped (Figure 1a), but one change at a time only the vocabulary edits did. Among open tools, aiseo-audit keeps an evidence tier per factor and demotes lists and tables to diagnostics [1]; we adapt code from it.

6 LIMITATIONS AND FUTURE WORK

Our one-change tests put the page among four fixed sources, so they measure citation, not whether the page is found; with retrieval back in such gains can reverse [4], and live engines differ and drift [6]. The effects are small, about 0.03 on a 0-to-1 score on which the unedited page averages 0.54, or about one rank position over eight answers. The keyword in our last test is two words of the real query, so it carries part of the query by design; real keywords may share less. We followed up the question section because it passed on our first test's queries, so estimates that include them may run high; the 180 new queries avoid this but are longer, and the gain rose with query length: at the first test's query lengths it is +0.033. An earlier sensitivity test missed its preset bar: a one-sentence answer raised the page's citation share by 0.007 to 0.034, not the 0.10 required; our judges come from its exploratory part. The tests cover 320 English queries, nearly all edited by one model, from the same family as one of the judges. The planned byte check of the harness prompt against the bench's has not been run. Next are a comparison with live engines (a planned Gemini-judge check stalled on API quota) and a study with writers.

7 ETHICS AND USE OF DATA

No human participants took part and no user data was collected; a future user study will go through IRB review. Sample drafts and competitor pages are invented; C-SEO Bench is downloaded at run time, never redistributed; building competitor pages from URLs respects robots.txt. On dual use, the tool writes no page text on its own, asks for real and checkable material, and states the FTC's endorsement rules; the bench could test a manipulative edit, but the tool offers none. The tests use the public pages GEO-bench lists [2]; their text, and everything built from it, is not redistributed, only URLs, hashes and results.

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Use of generative AI. We used Claude (Anthropic) as an assistant for software development, experiment and analysis code, and writing and editing. The authors designed the study, checked the results and take full responsibility for the paper.

REFERENCES

[1] AE Studio. 2026. *aiseo-audit, version 2.1.2 (commit 6c888aa).* Retrieved 2026-10-24 from https://github.com/agencyenterprise/aiseo-audit

[2] Pranjal Aggarwal, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, Karthik Narasimhan, and Ameet Deshpande. 2024. GEO: Generative Engine Optimization. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (Barcelona, Spain) *(KDD '24).* ACM, New York, NY, USA, 5–16. https://doi.org/10.1145/3637528.3671900 arXiv:2311.09735 [cs.LG]

[3] Fiona Draxler, Anna Werner, Florian Lehmann, Matthias Hoppe, Albrecht Schmidt, Daniel Buschek, and Robin Welsch. 2024. The AI Ghostwriter Effect: When Users do not Perceive Ownership of AI-Generated Text but Self-Declare as Authors. *ACM Transactions on Computer-Human Interaction* 31, 2 (2024), 1–40. https://doi.org/10.1145/3637875

[4] Sunghwan Kim, Wooseok Jeong, Serin Kim, Sangam Lee, and Dongha Lee. 2026. SAGEO Arena: A Realistic Environment for Evaluating Search-Augmented Generative Engine Optimization. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26).* ACM, New York, NY, USA, 2342–2353. https://doi.org/10.1145/3770855.3818146 arXiv:2602.12187 [cs.IR]

[5] Martin Lou, Jackie Crowley, Samuel Dodson, and Dongwook Yoon. 2025. AnchoredAI: Contextual Anchoring of AI Comments Improves Writer Agency and Ownership. arXiv:2509.16128 [cs.HC]

[6] Olivier Martinez. 2026. Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization (2023–2026). arXiv:2607.14035 [cs.IR]

[7] Haritz Puerto, Martin Gubri, Tommaso Green, Seong Joon Oh, and Sangdoo Yun. 2025. C-SEO Bench: Does Conversational SEO Work?. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track.* OpenReview.net. arXiv:2506.11097 [cs.CL]

[8] Rahul Vishwakarma, Shushant Kumar, and Ratnesh Jamidar. 2026. What Gets Cited: Competitive GEO in AI Answer Engines. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26).* ACM, New York, NY, USA, 4950–4954. https://doi.org/10.1145/3805712.3808445 arXiv:2605.25517 [cs.AI]

[9] Alexander Wan, Eric Wallace, and Dan Klein. 2024. What Evidence Do Language Models Find Convincing?. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers).* Association for Computational Linguistics, Bangkok, Thailand, 7468–7484. https://doi.org/10.18653/v1/2024.acl-long.403

[10] Stephanie T. Wang, Jeffrey Gleason, Yakov Bart, Christo Wilson, and Danaé Metaxa. 2026. AI in Search Reduces Publisher Referrals Without Improving User Experience: Experimental Evidence. arXiv:2608.18352 [cs.IR]

[11] Junwei Yu, Mufeng Yang, Yepeng Ding, and Hiroyuki Sato. 2026. Structural Feature Engineering for Generative Engine Optimization: How Content Structure Shapes Citation Behavior. arXiv:2603.29979 [cs.CL]